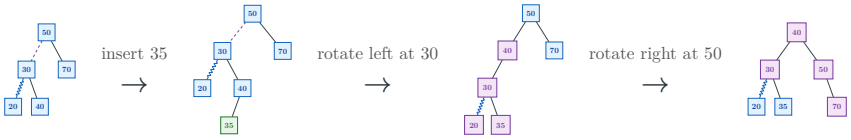
typed-dsa showcase

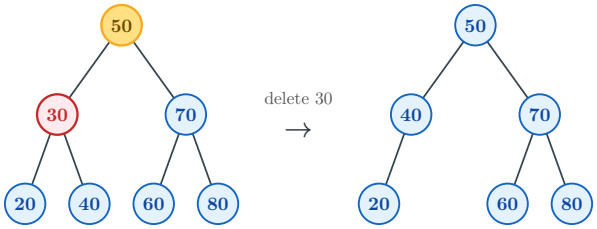
A deliberately over-styled pass through the package's main diagram families.

Trees and AVL transitions

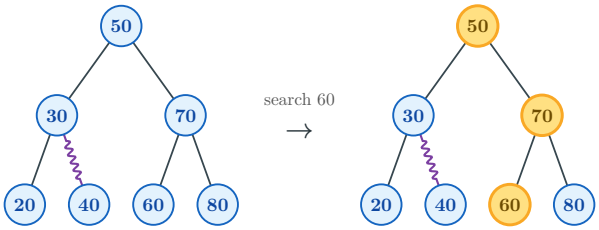
```
#let a = avl(50, 30, 70, 20, 40,
  style: your-style)
#(a.insert)(35, rebalance: (enabled:
true, all-steps: true)).diagram
```




```
#let b = bst(50, 30, 70, 20, 40, 60, 80,
  style: your-style)
#(b.delete)(30).diagram
```

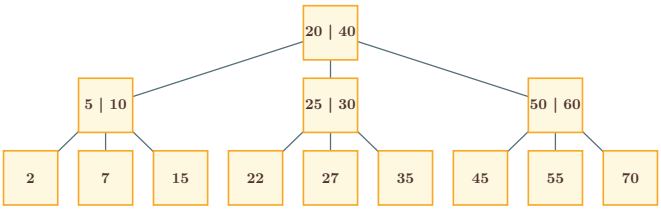



```
#let c = bst(50, 30, 70, 20, 40, 60, 80,
  style: your-style)
#(c.search)(60).diagram
```

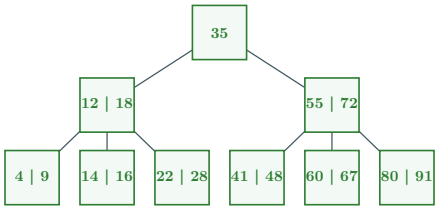


Multiway trees and table-like structures

```
#tree(
  node([20 | 40], children: (
    node([5 | 10], children: (...)),
    node([25 | 30], children: (...)),
    node([50 | 60], children: (...)),
  )),
  style: your-style,
)
```




```
#tree(
  node([35], children: (
    node([12 | 18], children: (...)),
    node([55 | 72], children: (...)),
  )),
  style: your-style,
)
```



Arrays and matrices

```
#array-view(
  8, 13, 21, 34, 55, 89, 144, 233, 377,
  style: your-style + (indices: (enabled:
true)),
  cell-customizations: ((4, pivot-style),
(5, pivot-style)),
)
```

8	13	21	34	55	89	144	233	377
0	1	2	3	4	5	6	7	8

```
#matrix(
  ((0, 4, 8, 12, 16), ...),
  row-labels: ([A], [B], [C], [D], [E]),
  column-labels: ([A], [B], [C], [D],
[E]),
  style: your-style,
)
```

	A	B	C	D	E
A	0	4	8	12	16
B	3	0	5	9	13
C	7	2	0	6	10
D	11	6	1	0	4
E	15	10	5	2	0

Sorting algorithms: pseudocode and traces

Each trace uses a small array and shows every generated step.

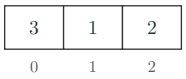
Bubble sort

BUBBLE SORT():

```
1  for p ← 0 to n − 2:
2      for j ← 0 to n − p − 2:
3          if A_j > A_{j+1}:
4              swap A_j and A_{j+1}
```

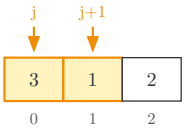
Step 1

start



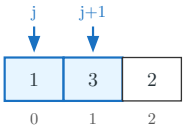
Step 2

compare 3 and 1



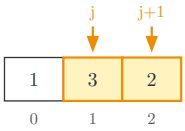
Step 3

swap 3 and 1



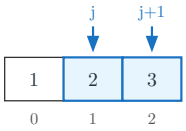
Step 4

compare 3 and 2



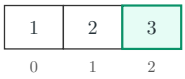
Step 5

swap 3 and 2



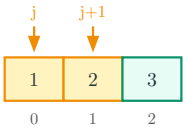
Step 6

3 settled



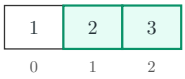
Step 7

compare 1 and 2



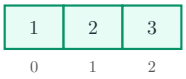
Step 8

2 settled



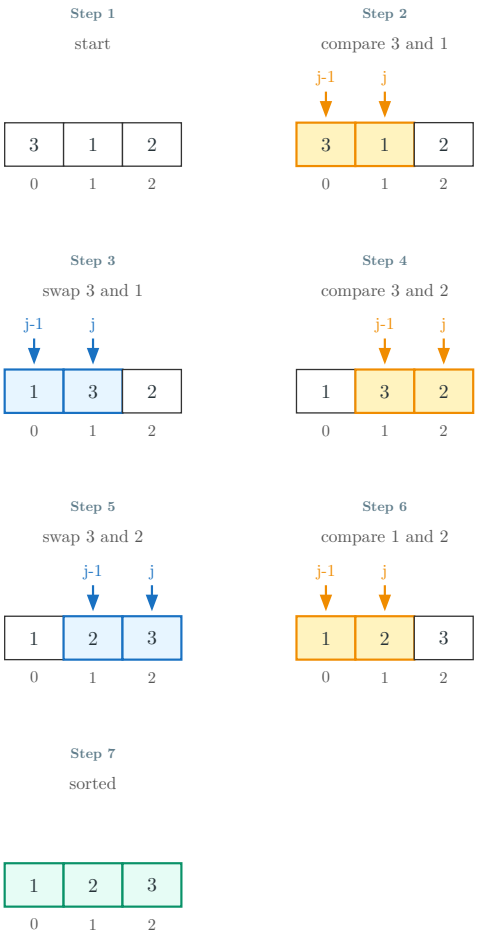
Step 9

sorted



Insertion sort

```
INSERTION SORT():
1  for i ← 1 to n − 1:
2      k ← A[i]
3      j ← i
4      while j > 0 and A[j] < A[j − 1]:
5          swap A[j] and A[j − 1]
6          j ← j − 1
```

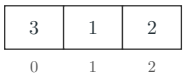


Selection sort

```
SELECTION SORT():
1  for i ← 0 to n − 1:
2      m ← i
3      for j ← i + 1 to n − 1:
4          if A[j] < A[m]:
5              m ← j
6      swap A[i] and A[m]
```

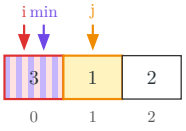
Step 1

start



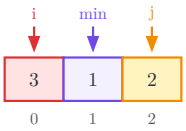
Step 2

position 0, minimum 0, item 1



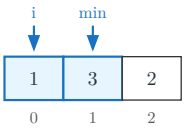
Step 3

position 0, minimum 1, item 2



Step 4

swap 3 and 1

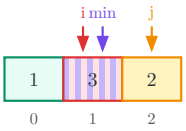


Step 5

1 settled

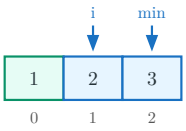
Step 6

position 1, minimum 1, item 2



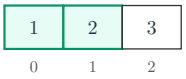
Step 7

swap 3 and 2



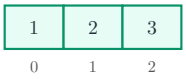
Step 8

2 settled



Step 9

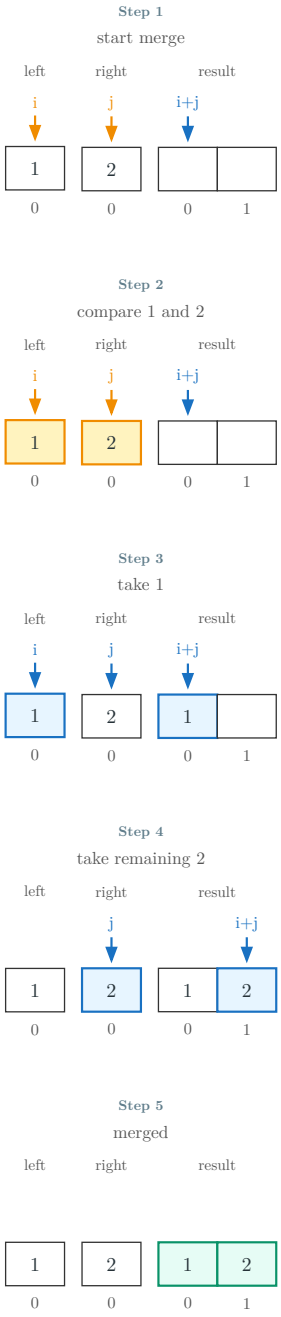
sorted



Merge operation

MERGE():

```
1  i ← 0; j ← 0; M ← ()
2  while i < |L| and j < |R|:
3    if L_i ≤ R_j:
4      append L_i to M; i ← i + 1
5    else:
6      append R_j to M; j ← j + 1
7  append remaining L and R to M
8  return M
```

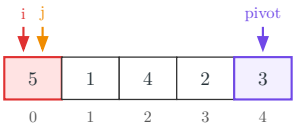


Quick sort: last-pivot partition

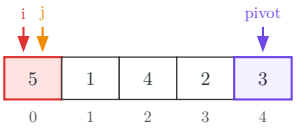
LAST-PIVOT PARTITION():

```
1  p ← A_{n−1}
2  i ← 0
3  for j ← 0 to n − 2:
4     if A_j ≤ p:
5        swap A_i and A_j
6        i ← i + 1
7  swap A_i and A_{n−1}
```

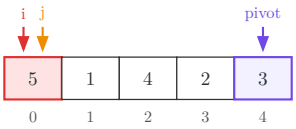
Step 1
start
pivot: 3 at 4 i: 0 j: 0



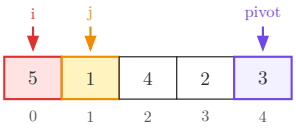
Step 2
select last pivot 3
pivot: 3 at 4 i: 0 j: 0



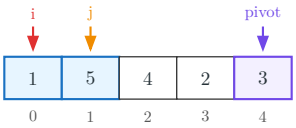
Step 3
compare 5 and pivot 3
pivot: 3 at 4 i: 0 j: 0



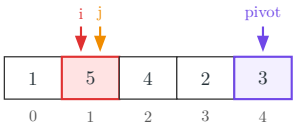
Step 4
compare 1 and pivot 3
pivot: 3 at 4 i: 0 j: 1



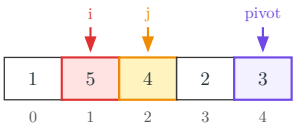
Step 5
swap 5 and 1
pivot: 3 at 4 i: 0 j: 1



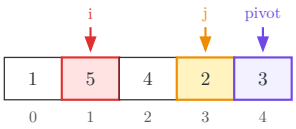
Step 6
advance i to 1
pivot: 3 at 4 i: 1 j: 1



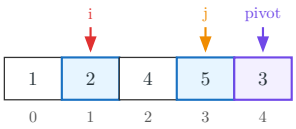
Step 7
compare 4 and pivot 3
pivot: 3 at 4 i: 1 j: 2



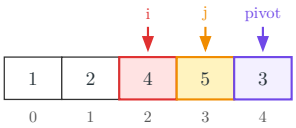
Step 8
compare 2 and pivot 3
pivot: 3 at 4 i: 1 j: 3



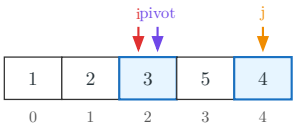
Step 9
swap 5 and 2
pivot: 3 at 4 i: 1 j: 3



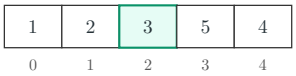
Step 10
advance i to 2
pivot: 3 at 4 i: 2 j: 3



Step 11
place pivot 3
pivot: 3 at 2 i: 2 j: 4

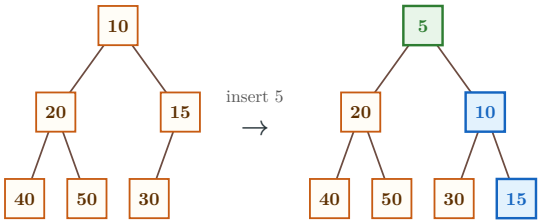


Step 12
partitioned
pivot: 3 at 2 i: 2 j: 4

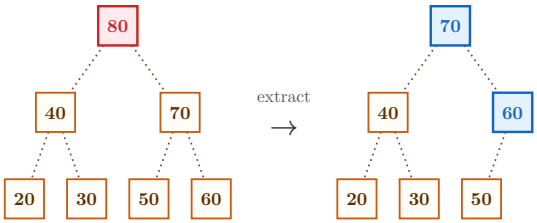


Heaps with sift paths

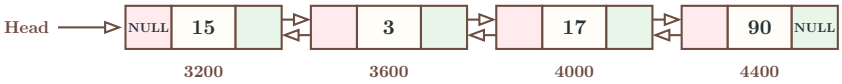
```
#let h = min-heap(10, 20, 15, 40, 50, 30,
  style: your-style)
#(h.insert)(5).diagram
```



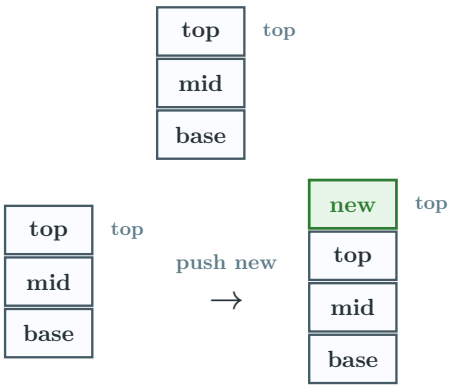

```
#let h = max-heap(50, 30, 70, 20, 40, 60,
80,
  style: your-style)
#(h.extract)().diagram
```

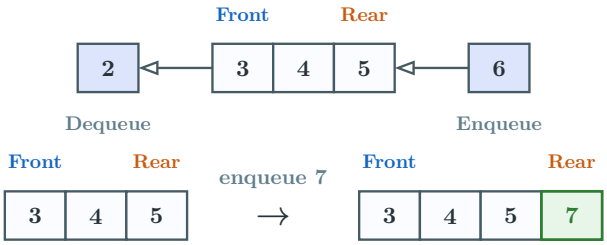


Linked structures

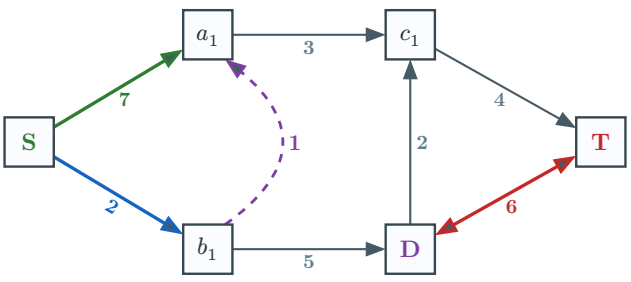


Stack and queue workflows





Weighted graph with custom edge labels



Hand-composed rotation schematic